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If there are 13 boys in club X, what is the average age of these boys?

1. The oldest boy is 13 years old and the youngest boy is 9 years old.
2. Eleven of the boys are either 10 years old or 11 years old.

Arithmetic Statistics

What is the average age of the 13 boys, or equivalently, what is the sum of the ages of the 13 boys?

1. Given that the oldest boy is 13 and the youngest boy is 9, it is not possible to determine the sum of their ages. For example, if their ages were such that one is 9, eleven are 10, and one is 13, then the sum of their ages would be less than if their ages were such that one is 9, eleven are 11, and one is 13; NOT sufficient.
2. Given that eleven of the boys are either 10 or 11, it is not possible to determine the sum of their ages, because the same examples used above are such that eleven of the boys are either 10 or 11; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the sum of their ages because the examples above satisfy both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

•

If all the employees of a company fall into one and only one of 3 groups, X, Y, or Z, with 250, 100, and 20 members in each group, respectively, what is the average (arithmetic mean) weekly salary of all the employees of this company, if all employees are paid every week of the year?

1. The average (arithmetic mean) annual salary of the employees in Group X is $10,000, in Group Y $15,000 and in Group Z $20,000.
2. The total annual payroll is $4,400,000.

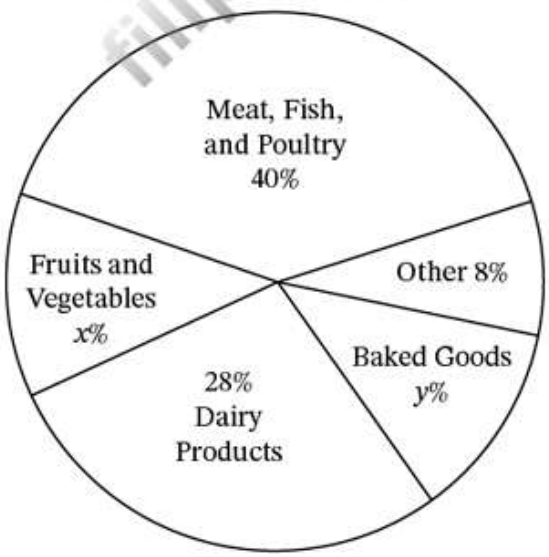
Arithmetic Statistics

The average of the weekly salaries is the average of the annual salaries divided by 52, and thus the average of the weekly salaries can be determined if and only if the average of the annual salaries can be determined. What is the average of the annual salaries, or equivalently, what is the sum of the annual salaries?

1. Given that the average of the annual salaries of employees in Groups X, Y, and Z is $10,000, $15,000, and $20,000, respectively, it follows that the sum of the annual salaries of employees in Groups X, Y, and Z is 250($10,000), 100($15,000), and 20($20,000), respectively. Therefore, the sum of the annual salaries is the sum of these three amounts; SUFFICIENT.
2. We are given that the sum of the annual salaries is $4,400,000; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

DISTRIBUTION OF SALES INCOME
FOR STORE *S* LAST WEEK



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$100\% = \$100{,}000$

•

According to the graph above, the sale of fruits and vegetables in Store S last week accounted for what percent of the total sales income for the week?

1. Last week the total income from the sale of fruits and vegetables in Store S was $16,000.
2. $x = 2y$

Algebra Percents

According to the graph, sales of fruits and vegetables accounted for $x\%$ of the total sales, baked goods accounted for $y\%$, and all other categories combined accounted for 76%. It follows that $x + y = 100 - 76 = 24$. Determine the value of x.

1. Given that sales of fruits and vegetables accounted for $x\%$ of the total sales of $100,000 and this amounted to $16,000, it follows that $\frac{x}{100}(100{,}000) = 16{,}000$. Thus, $x = 16$; SUFFICIENT.
2. Given that $x = 2y$, it follows that $x + y = 3y = 24$, $y = 8$, and $x = 16$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

Larry saves x dollars per month. Will Larry's total savings one year from now exceed his present savings by at least $500? (Assume that there is no interest.)

1. In 6 months Larry's total savings will be $900.
2. In 3 months Larry's total savings will exceed his present savings by $150.

Algebra Applied Problems

Let $\$p$ be Larry's present savings. Larry saves $\$x$ per month. One year from now (i.e., 12 months from now), Larry will have saved an additional $\$12x$. Determine if $(p + 12x) - p > 500$ or, equivalently, determine if $12x > 500$.

1. Given that in 6 months Larry will have saved a total of $900, it follows that $p + 6x = 900$, but since the value of p is unknown, it cannot be determined if $12x > 500$; NOT sufficient.
2. Given that $(p + 3x) - p = \$150$, it follows that $x = 50$, $12x = 600$, and $12x > 500$; SUFFICIENT.

The correct answer is B;statement 2 alone is sufficient.

•

If Randy has twice as many coins as Alice, and if Maria has 7 times as many coins as Alice, what is the combined number of coins that all three of them have?

1. Alice has 4 fewer coins than Randy.
2. Maria has 20 more coins than Randy.

Algebra Simultaneous Equations

Determine the total number of coins that Randy, Alice, and Maria have, given that Randy has twice as many coins as Alice and Maria has 7 times as many coins as Alice. In other words, determine the value of $R + A + M$, where R, A, and M represent the number of coins, respectively, that Randy, Alice, and Maria have, given that $R = 2A$ and $M = 7A$.

1. Given that Alice has 4 fewer coins than Randy, it follows that $A = R - 4$. Since $R = 2A$, it follows that $A = 2A - 4$, from which the value of A can be determined. From the value of A, values for R and M can be determined and $R + A + M$ can also be determined; SUFFICIENT.
2. Given that Maria has 20 more coins than Randy, it follows that $M = R + 20$. Since $M = 7A$ and $R = 2A$, it follows that $7A = 2A + 20$, from which the value of A can be determined. From the value of A, values for R and M can be determined and $R + A + M$ can also be determined; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

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•

A line of people waiting to enter a theater consists of seven separate and successive groups. The first person in each group purchases one ticket for each person in the group and for no one else. If n is the total number of tickets sold for the first six groups, is n an even number?

1. There are no more than 4 people in each group.
2. The 19th person in line purchases the tickets for the seventh group.

Arithmetic Applied Problems

Determine whether the total number of people in the first six groups is an even number.

1. Given that each group contains at most 4 people, it is not possible to determine whether the total number of people in the first six groups is an even number. For example, if the numbers of people in the first six groups were 2, 2, 2, 2, 2, and 2, then the total number of people in the first six groups would be 12, which is an even number. However, if the numbers of people in the first six groups were 2, 2, 2, 2, 2, and 3, then the total number of people in the first six groups would be 13, which is not an even number; NOT sufficient.
2. Given that the 19th person in line purchased the tickets for the seventh group, it follows that the total number of people in the first six groups was 18, which is an even number; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

If John has exactly 10 coins each of which was minted in 1910 or 1920 or 1930, how many of his coins were minted in 1920?

1. Exactly 6 of his coins were minted in 1910 or 1920.
2. Exactly 7 of his coins were minted in 1920 or 1930.

Algebra Simultaneous Equations

Let x, y, and z be the numbers of coins John has that were minted in, respectively, 1910, 1920, and 1930. Then x, y, and z are nonnegative integers and $x + y + z = 10$. What is the value of y?

1. Given that $x + y = 6$, it is possible that $y = 2$ (e.g., $x = 4$, $y = 2$, and $z = 4$) and it is possible that $y = 4$ (e.g., $x = 2$, $y = 4$, and $z = 4$); NOT sufficient.
2. Given that $y + z = 7$, it is possible that $y = 3$ (e.g., $x = 3$, $y = 3$, and $z = 4$) and it is possible that $y = 4$ (e.g., $x = 3$, $y = 4$, and $z = 3$); NOT sufficient.

Taking (1) and (2) together, from $x + y + z = 10$ and (1) it follows that $6 + z = 10$, or $z = 4$. Substituting $z = 4$ into $y + z = 7$ gives $y + 4 = 7$, or $y = 3$.

The correct answer is C; both statements together are sufficient.

•

The total profit of corporation K was $3,400,000 in year X. What was the total profit in year Y?

1. Income in year Y was 30 percent more than in year X.
2. Costs in year Y were 40 percent more than in year X.

Algebra Applied Problems

Let $\$C_X$ be costs in year X, $\$I_X$ be income in year X, $\$C_Y$ be costs in year Y, and $\$I_Y$ be income in year Y. Given that $I_X - C_X = 3{,}400{,}000$, determine the value of $I_Y - C_Y$.

1. Given that $I_Y = 1.3I_X$, some information about income in year Y is known, but since nothing is known about costs in year Y, the profit in year Y cannot be determined; NOT sufficient.
2. Given that $C_Y = 1.4C_X$, some information about costs in year Y is known, but since nothing is known about income in year Y, the profit in year Y cannot be determined; NOT sufficient.

Taking (1) and (2) together,
$I_Y - C_Y = 1.3I_X - 1.4C_X = 1.3(I_X - C_X) - 0.1C_X = 1.3(3{,}400{,}000) - 0.1C_X$, which can have more than one possible value. The table below gives two specific examples that illustrate this.

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than one possible value. The table below gives two specific examples that illustrate this.

C_X	1,000,000	2,000,000
I_X	4,400,000	5,400,000
$I_X - C_X$	3,400,000	3,400,000
$C_Y = 1.4C_X$	1,400,000	2,800,000
$I_Y = 1.3I_X$	5,720,000	7,020,000
$I_Y - C_Y$	4,320,000	4,220,000

The correct answer is E; both statements together are still not sufficient.

•

Zelma scored 90, 88, and 92 on 3 of the 6 mathematics tests that she took. What was her average (arithmetic mean) score on the 6 tests?

1. Her average (arithmetic mean) score on 5 of the tests was 90.
2. Her score on one of the tests was 91.

Arithmetic Statistics

Determine the average (arithmetic mean) of Zelma's 6 test scores, given that 3 of the scores are 90, 88, and 92.

1. Given that the average score on 5 of the 6 tests was 90, it is not possible to determine the average of all 6 test scores. For example, the 6 test scores could be 89, 91, 90, 88, 92, and 100 for an average score of $91\frac{2}{3}$ or the 6 test scores could be 89, 91, 90, 88, 92, and 50 for an average score of $83\frac{1}{3}$; NOT sufficient.
2. Given that one of the scores was 91, it is not possible to determine the average of all 6 test scores. For example, the 6 test scores could be 89, 91, 90, 88, 92, and 100 for an average score of $91\frac{2}{3}$ or the 6 test scores could be 89, 91, 90, 88, 92, and 50 for an average score of $88\frac{1}{3}$; NOT sufficient.

Taking (1) and (2) together and noting that the examples that were used to show that (2) is not sufficient were the same examples that were used to show that (1) is not sufficient, the average of Zelma's 6 test scores cannot be determined.

The correct answer is E; both statements together are still not sufficient.

•

Will a company lose sales if its strategy focuses on consumer well-being rather than exploiting consumer fears?

1. The company's current marketing strategy exploits its consumers' fears to drive sales, resulting in increased profits.
2. From an ethical standpoint, the company should prioritize consumer well-being over short-term profits and refrain from exploiting their fears.

Inference

1. Even if the company's current fear-based marketing strategy has increased sales and profits, a new strategy focused on consumer well-being might work even better. Thus, whether the company will lose sales if its strategy focuses on consumer well-being rather than exploiting consumer feats cannot be determined from (1) alone; NOT sufficient.
2. What the company should do from an ethical standpoint is irrelevant to whether any marketing strategy will

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make the company lose sales. Thus, whether the company will lose sales if its strategy focuses on consumer well-being rather than exploiting consumer feats cannot be determined from (2) alone; NOT sufficient.

Since (1) alone is insufficient to answer the question, and (2) is completely irrelevant to answering it, both (1) and (2) together are insufficient to answer it. Whether or not the company's current fear-based marketing strategy has been profitable, and whether or not that strategy is ethical, a new strategy focused on consumer well-being might or might not make the company lose sales.

The correct answer is E; both statements together are still not sufficient.

•

Should a holistic approach be used to measure academic success?

1. Traditional education policies focus on standardized testing as the primary measure of academic success.
2. To foster holistic learning, education policies should prioritize a well-rounded curriculum that includes critical thinking, creativity, and practical skills.

Inference

1. The observation that traditional education policies focus on standardized testing to measure academic success tells us nothing about whether or not this traditional approach is better than, worse than, or even distinct from a "holistic" approach. Thus, whether a holistic approach should be used to measure academic success cannot be determined from (1) alone; NOT sufficient.
2. Even if education policies should prioritize a well-rounded curriculum to foster holistic learning, a holistic approach might not be the best way to measure academic success. Thus, whether a holistic approach should be used to measure academic success cannot be determined from (2) alone; NOT sufficient.

Now suppose traditional education policies measure academic success with standardized testing as (1) says, while education policies should prioritize a well-rounded curriculum to foster holistic learning as (2) says. In that case, a holistic approach to measuring academic success might or might not be better than, worse than, or even distinct from the traditional approach.

The correct answer is E; both statements together are still not sufficient.

•

Can financial budgeting help individuals avoid debt?

1. Many individuals struggle with mounting debt due to excessive spending resulting from a lack of financial planning.
2. To avoid taking out loans, individuals should practice responsible budgeting, enabling them to save for emergencies and achieve financial stability.

Inference

1. The claim that many individuals struggle with mounting debt due to excessive spending resulting from lack of financial planning implies that they might not struggle with mounting debt if they *did* use financial planning to avoid excessive spending; that is, if they practiced financial budgeting. In other words, (1) implies that financial budgeting could help these individuals avoid the mounting debt that they struggle with. Thus, (1) alone entails that financial budgeting can help in avoiding debt; SUFFICIENT.
2. The claim that individuals should practice responsible budgeting in order to avoid taking out loans implies that responsible budgeting can help them avoid taking out loans. Assuming that in at least some of these cases, avoiding taking out a loan can allow one to avoid the debt from that loan, this means that responsible financial budgeting can help in avoiding debt. Thus, (2) alone entails that financial budgeting can help in avoiding debt; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

What kind of behavior is usually displayed by conflicting factions?

1. In conflicts, some factions resort to aggression and hostility, which escalates the situation.
2. To promote peaceful resolutions, factions should engage in open communication, empathy, and compromise to find mutually beneficial solutions.

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Inference

1. Even if some factions in conflicts resort to hostility and aggression, conflicting factions might not usually display such behavior. Thus, what kind of behavior conflicting factions usually display cannot be determined from (1) alone; NOT sufficient.
2. A claim about what conflicting factions should do in order to promote peaceful resolutions doesn't clearly indicate what they usually do in reality. Thus, what kind of behavior conflicting factions usually display cannot be determined from (2) alone; NOT sufficient.

(1)'s observation that some conflicting factions resort to hostility and aggression, even together with (2)'s claim that such factions should behave differently to promote peaceful resolutions, still doesn't tell us how conflicting factions usually behave.

The correct answer is E; both statements together are still not sufficient.

•

What is the probability of landing on tails at least once in a sequence of 100 tosses of a specific coin?

1. When tossing the coin, there are only two possible outcomes, heads or tails, and both have an equal chance of occurring.
2. In a series of 100 tosses of the coin, we can expect almost 50 heads.

Evaluate

1. Since (1) tells us that there are only two possible outcomes to tossing the coin, that each of those outcomes has a 50% chance of occurring, and that one of them is tails, we can calculate the chance of landing on tails at least once in a series of 100 tosses of the coin as 1 − (0.5)^100. Thus, (1) alone lets us determine this probability; SUFFICIENT.
2. The vague thought that "we can expect almost 50 heads" in a series of 100 tosses of the coin doesn't tell us the probability of the coin landing on tails with each toss, so it doesn't give us enough information to calculate the probability of landing on tails at least once in a series of 100 tosses. Thus, (2) alone doesn't let us determine this probability. NOT SUFFICIENT.

The correct answer is A; statement 1 alone is sufficient.

•

What are some possible signs of impending rainfall in Exville today?

1. The weather forecast correctly predicts a 90% chance of rain in Exville today based on heavy cloud cover and low-pressure systems moving into the region.
2. Given the darkening sky, the sudden drop in temperature, and strong winds, it is highly likely that a rainstorm is imminent in Exville today.

Inference

1. If a weather forecast is correct in predicting a 90% chance of rain in a town because of heavy cloud cover and low-pressure systems moving into the region, then heavy cloud cover and low-pressure systems must be two of the possible signs of impending rainfall in that town. Thus, (1) alone provides examples of some possible signs of impending rainfall in Exville today; SUFFICIENT.
2. If a darkening sky, a sudden drop in temperature, and strong winds are reasons why it is highly likely that a rainstorm is imminent in Exville today, then those factors must be three of the possible signs of impending rainfall in Exville today. Thus, (2) alone provides examples of some possible signs of impending rainfall in Exville today; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

A patient experiencing cold-like symptoms goes to a doctor. Is the patient suffering from a viral infection?

1. Based on their symptoms and medical history, the patient thinks they are suffering from a viral infection.
2. Due to their knowledge of common illnesses in the local area, the doctor thinks it is more likely that seasonal allergies are the cause of the patient's symptoms.

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Inference

1. Even if the patient's symptoms and medical history give the patient reason to believe that a viral infection is causing the symptoms, that belief may be mistaken. Thus, whether the patient is suffering from a viral infection cannot be determined from (1) alone; NOT sufficient.
2. Even if the doctor reasonably believes that seasonal allergies are more likely than a viral infection to be causing the patient's symptoms, a viral infection might still be the cause. Thus, whether the patient is suffering from a viral infection cannot be determined from (2) alone; NOT sufficient.

If (1) and (2) are both true, the patient and the doctor have opposing beliefs about the likelihood that the patient has a viral infection. These opposing beliefs don't settle the question of whether the patient has such an infection or not.

The correct answer is E; both statements together are still not sufficient.

•

What should generally be expected if you venture into areas surrounding a road closure?

1. With a major G20 event happening at the Jawahar Lal Nehru Stadium tonight and reports of road closures in the area, there's a strong likelihood of congestion.
2. Given the large number of attendees expected at an event at the Jawahar Lal Nehru Stadium tonight, shuttle buses will be provided between the stadium and the International Media Centre.

Inference

1. Observations about specific conditions around a stadium on a specific night can't support any broad generalization about what to expect around areas surrounding road closures. Thus, what should generally be expected if you venture into areas surrounding a road closure cannot be determined from (1) alone; NOT sufficient.
2. Observations about specific conditions around a stadium on a specific night can't support any broad generalization about what to expect around areas surrounding road closures. Thus, what should generally be expected if you venture into areas surrounding a road closure cannot be determined from (2) alone; NOT sufficient.

(1) and (2) both describe conditions around the same stadium on a specific night. Thus, even both statements together can't support any broad generalization about what to expect around areas surrounding road closures.

The correct answer is E; both statements together are still not sufficient.

•

To fill an order on schedule, a manufacturer had to produce an average (arithmetic mean) of 1,000 tools per day for n days. What is the value of n?

1. Because of production problems, the manufacturer produced only 600 tools per day during the first 5 days.
2. Because of production problems, the manufacturer had to produce 1,500 tools per day on each of the last 4 days in order to meet the schedule.

Algebra Applied Problems

1. Given that 600 tools per day were produced during the first 5 days, which is a total of 3,000 tools, the value of n cannot be determined because there is no information about how many tools were produced in the later days. The number of later days needed is increased when the amount over 1,000 in the production of tools per day during these later days is decreased. For example, $n = 10$ is possible (5 remaining days in which 1,400 tools per day were produced; this results in 3,000 + 7,000 = 10,000 tools produced in 10 days), and $n = 15$ is possible (10 remaining days in which 1,200 tools per day were produced; this results in 3,000 + 12,000 = 15,000 tools produced in 15 days); NOT sufficient.
2. Given that 1,500 tools per day were produced during the last 4 days, which is a total of 6,000 tools, the value of n cannot be determined because there is no information about how many tools were produced in the earlier days. The number of earlier days needed is increased when the amount under 1,000 in the production of tools per day during these earlier days is decreased. For example, $n = 8$ is possible (4 earlier days in which 500 tools per day were produced; this results in 6,000 + 2,000 = 8,000 tools produced in 8 days), and $n = 12$ is possible (8 earlier days in which 750 tools per day were produced; this results in 6,000 + 6,000 = 12,000 tools produced in 12 days); NOT sufficient.

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Taking (1) and (2) together, a total of 3,000 + 6,000 = 9,000 tools were produced in the 9 days we are given information about. Therefore, $n = 9$ is possible. However, $n = 11$ is also possible by including, between the first 5 days and the last 4 days, a day in which the production was 900 tools and another day in which the production was 1,100 tools (i.e., an additional 2,000 tools produced during an additional 2 days).

The correct answer is E; both statements together are still not sufficient.

Questions 76 to 113 — Difficulty: Hard

76. The only types of vehicles sold at a certain dealership last month were sedans, trucks, and vans. If the ratio of the number of sedans to the number of trucks to the number of vans sold at the dealership last month was 4:5:7, respectively, what was the total number of vehicles sold at the dealership last month?

    1. The number of vans sold at the dealership last month was between 10 and 20.
    2. The number of sedans sold at the dealership last month was less than 10.

    Arithmetic Ratio and Proportion

    Letting k be the constant of proportionality, the number of sedans, trucks, and vans sold at the dealership last month was, respectively, $4k$, $5k$, and $7k$. Since $4k$ is a positive integer, k must be a fraction having denominator 1, 2, or 4 when reduced to lowest terms. Since $5k$ is a positive integer, k must be a fraction having denominator 1 or 5 when reduced to lowest terms. It follows that k must be a fraction having denominator 1 when reduced to lowest terms, and hence k is a positive integer. We are to determine the value of $4k + 5k + 7k = 16k$.

    1. Given that $10 < 7k < 20$ and k is a positive integer, it follows that $k = 2$. Therefore, the total number of vehicles sold at the dealership last month was $16k = 32$; SUFFICIENT.
    2. Given that $4k < 10$ and k is a positive integer, it follows that $k = 1$ or $k = 2$. Therefore, the total number of vehicles sold at the dealership last month could be $16k = 16$, and it could be $16k = 32$; NOT sufficient.

    The correct answer is A; statement 1 alone is sufficient.

77. Machine M, working alone at its constant rate, produces x widgets every 4 minutes. Machine N, working alone at its constant rate, produces y widgets every 5 minutes. If machines M and N work simultaneously at their respective constant rates for 20 minutes, does machine M produce more widgets than machine N in that time?

    1. $x > 0.8y$
    2. $y = x + 1$

    Algebra Applied Problems

    Since machine M produces x widgets every 4 minutes, machine M produced $5x$ widgets during the 20 minutes. Moreover, since machine N produces y widgets every 5 minutes, machine N produced $4y$ widgets during the 20 minutes. We are to determine whether $5x > 4y$.

    1. Given that $x > 0.8y$, multiplying both sides by 5 gives $5x > 4y$; SUFFICIENT.
    2. Given that $y = x + 1$, substituting $y = x + 1$ into $5x > 4y$ gives $5x > 4(x + 1)$, or $x > 4$, which can be used to obtain examples showing that it is not possible to determine whether $5x > 4y$. For example, if $x = 5$ and $y = 6$, then $y = x + 1$ and $5x > 4y$ (because 25 > 24). On the other hand, if $x = 3$ and $y = 4$, then $y = x + 1$ and $5x < 4y$ (because 15 < 16); NOT sufficient.

    The correct answer is A; statement 1 alone is sufficient.

78. What percent of the students at University X are enrolled in a science course but are not enrolled in a biology course?

    1. 28 percent of the students at University X are enrolled in a biology course.
    2. 70 percent of the students at University X who are enrolled in a science course are enrolled in a biology course.

    Algebra Percents

    Under the assumption that a biology course is a type of science course, determine the percent of University X students who are enrolled in a science course, but not in a biology course.

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1. Given that 28% of the students at University X are enrolled in a biology course, if 100% of the students are enrolled in a science course, then (100 − 28)% = 72% are enrolled in a science course, but not in a biology course. On the other hand if 50% of the students at University X are enrolled in a science course, then (50 − 28)% = 22% are enrolled in a science course, but not in a biology course; NOT sufficient.
2. Given that 70% of the students at University X who are enrolled in a science course are enrolled in a biology course, if 100% of the students at University X are enrolled in a science course, then (100 − 70)% = 30% are enrolled in a science course, but not in a biology course. On the other hand if 50% of the students at University X are enrolled in a science course, then 70% of 50% = 35% are enrolled in a biology course, (50 − 35)% = 15% are enrolled in a science course, but not in a biology course; NOT sufficient.

Taking (1) and (2) together, $0.28 = 0.7x$ where x is the percent of the students at University X who are enrolled in a science course. It follows that $x = 0.4$ or 40%. Thus, (40 − 28)% = 12% of the students at University X are enrolled in a science course, but not in a biology course.

The correct answer is C; both statements together are sufficient.

79. Each Type A machine fills 400 cans per minute, each Type B machine fills 600 cans per minute, and each Type C machine installs 2,400 lids per minute. A lid is installed on each can that is filled and on no can that is not filled. For a particular minute, what is the total number of machines working?

    1. A total of 4,800 cans are filled that minute.
    2. For that minute, there are 2 Type B machines working for every Type C machine working.

    Algebra Simultaneous Equations

    1. Given that 4,800 cans were filled that minute, it is possible that 12 Type A machines, no Type B machines, and 2 Type C machines were working, for a total of 14 machines, since (12)(400) + (0)(600) = 4,800 and (2)(2,400) = 4,800. However, it is also possible that no Type A machines, 8 Type B machines, and 2 Type C machines were working, for a total of 10 machines, since (0)(400) + (8)(600) = 4,800 and (2)(2,400) = 4,800; NOT sufficient.
    2. Given that there are 2 Type B machines working for every Type C machine working, it is possible that there are 6 machines working—3 Type A machines, 2 Type B machines, and 1 Type C machine. This gives 3(400) + 2(600) = 2,400 cans and 1(2,400) = 2,400 lids. It is also possible that there are 12 machines working—6 Type A machines, 4 Type B machines, and 2 Type C machines. This gives 6(400) + 4(600) = 4,800 cans and 2(2,400) = 4,800 lids; NOT sufficient.

    Taking (1) and (2) together, since there were 4,800 cans filled that minute, there were 4,800 lids installed that minute. It follows that 2 Type C machines were working that minute, since (2)(2,400) = 4,800. Since there were twice this number of Type B machines working that minute, it follows that 4 Type B machines were working that minute. These 4 Type B machines filled (4)(600) = 2,400 cans that minute, leaving 4,800 − 2,400 = 2,400 cans to be filled by Type A machines. Therefore, the number of Type A machines working that minute was $\frac{2{,}400}{400} = 6$, and it follows that the total number of machines working that minute was 2 + 4 + 6 = 12.

    The correct answer is C; both statements together are sufficient.

80. In a two-month survey of shoppers, each shopper bought one of two brands of detergent, X or Y, in the first month and again bought one of these brands in the second month. In the survey, 90 percent of the shoppers who bought Brand X in the first month bought Brand X again in the second month, while 60 percent of the shoppers who bought Brand Y in the first month bought Brand Y again in the second month. What percent of the shoppers bought Brand Y in the second month?

    1. In the first month, 50 percent of the shoppers bought Brand X.
    2. The total number of shoppers surveyed was 5,000.

    Arithmetic Percents

    This problem can be solved by using the following contingency table where A and B represent, respectively, the number of shoppers who bought Brand X and the number of shoppers who bought Brand Y in the first month; C and D represent, respectively, the number of shoppers who bought Brand X and the number of shoppers who bought Brand Y in the second month; and T represents the total number of shoppers in the survey. Also in the table, $0.9A$ represents the 90% of the shoppers who bought Brand X in the first month and also bought it in the second month, and $0.1A$ represents the (100 − 90)% = 10% of the shoppers who bought Brand X in the first

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second month, and $0.1A$ represents the (100 − 90)% = 10% of the shoppers who bought Brand X in the first month and Brand Y in the second month. Similarly, $0.6B$ represents the 60% of the shoppers who bought Brand Y in the first month and also bought it in the second month, and $0.4B$ represents the (100 − 60)% = 40% of the shoppers who bought Brand Y in the first month and Brand X in the second month.

		Second Month		
		X	**Y**	**Total**
First Month	**X**	$0.9A$	$0.1A$	A
	Y	$0.4B$	$0.6B$	B
	Total	C	D	T

Determine the value of $\frac{D}{T}$ as a percentage.

1. This indicates that 50% of the shoppers bought Brand X in the first month, so $A = 0.5T$. It follows that the other 50% of the shoppers bought Brand Y in the first month, so $B = 0.5T$. Then, $D = 0.1A + 0.6B = 0.1(0.5T) + 0.6(0.5T) = 0.05T + 0.30T = 0.35T$. It follows that $\frac{D}{T} = \frac{0.35T}{T} = 0.35$, which is 35%; SUFFICIENT.
2. This indicates that $T = 5{,}000$, as shown in the following table:

		Second Month		
		X	**Y**	**Total**
First Month	**X**	$0.9A$	$0.1A$	A
	Y	$0.4B$	$0.6B$	B
	Total	C	D	5,000

But not enough information is given to be able to determine D or D as a percentage of 5,000; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

81. If the total price of n equally priced shares of a certain stock was $12,000, what was the price per share of the stock?

1. If the price per share of the stock had been $1 more, the total price of the n shares would have been $300 more.
2. If the price per share of the stock had been $2 less, the total price of the n shares would have been 5 percent less.

Arithmetic Arithmetic Operations; Percents

Since the price per share of the stock can be expressed as $\frac{\$12{,}000}{n}$, determining the value of n is sufficient to answer this question.

1. A per-share increase of $1 and a total increase of $300 for n shares of stock mean together that $n(\$1) = \$300$. It follows that $n = 300$; SUFFICIENT.
2. If the price of each of the n shares had been reduced by $2, the total reduction in price would have been 5

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2. If the price of each of the n shares had been reduced by \$2, the total reduction in price would have been 5 percent less or 0.05(\$12,000). The equation $2n = 0.05(\$12{,}000)$ expresses this relationship. The value of n can be determined to be 300 from this equation; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

82. In Year X, 8.7 percent of the men in the labor force were unemployed in June compared with 8.4 percent in May. If the number of men in the labor force was the same for both months, how many men were unemployed in June of that year?

    1. In May of Year X, the number of unemployed men in the labor force was 3.36 million.
    2. In Year X, 120,000 more men in the labor force were unemployed in June than in May.

    Arithmetic Percents

    Since 8.7 percent of the men in the labor force were unemployed in June, the number of unemployed men could be calculated if the total number of men in the labor force was known. Let t represent the total number of men in the labor force.

    1. This implies that for May $(8.4\%)t = 3{,}360{,}000$, from which the value of t can be determined; SUFFICIENT.
    2. This implies that $(8.7\% - 8.4\%)t = 120{,}000$ or $(0.3\%)t = 120{,}000$. This equation can be solved for t; SUFFICIENT.

    The correct answer is D; each statement alone is sufficient.

83. On Monday morning a certain machine ran continuously at a uniform rate to fill a production order. At what time did it completely fill the order that morning?

    1. The machine began filling the order at 9:30 a.m.
    2. The machine had filled $\frac{1}{2}$ of the order by 10:30 a.m. and $\frac{5}{6}$ of the order by 11:10 a.m.

    Arithmetic Arithmetic Operations

    1. This merely states what time the machine began filling the order; NOT sufficient.
    2.

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